

Chapter

6

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(1)

Probability and Probability Distributions

Introduction to Probability

6

(1)

6.1 Introduction to Probability (Definition of terms)

Definition 1:- In statistics, any process of observation or measurement is called an **experiment.**

Example 1:

- i. To count the number of times a head appears on top in tossing a coin five times.
- ii. To identify the number of defective items from a set of items.

Tossing (or flipping) a coin





Head



Tail

The terms "heads" and "tails" on coins originated from the images on ancient coins and the terminology used in games of chance: The front of Athenian coins featured the head of Athena, and the back featured her sacred animal, the owl hanging down.



Head



Tail

6.1 Introduction to Probability

Definition of terms(Contd. . . .)



6.1 Introduction to Probability (Definition of terms)



(a)



(b)

6.1 Introduction to Probability

Definition of terms (Contd. . . .)

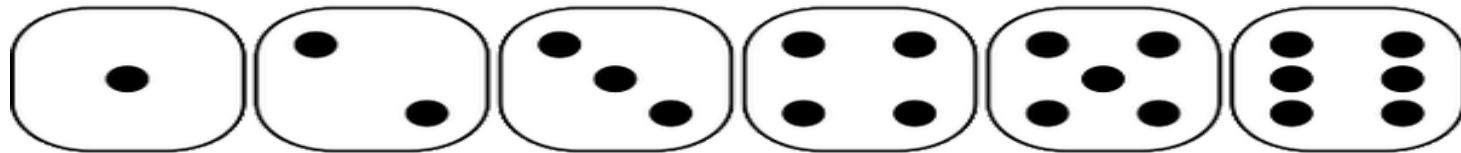
Definition 2:- The result we obtained from an experiment are called outcomes.

Example 2:- i. Count the total number of times a head appears on top in tossing a coin three times. Here the possible outcomes are 0, 1, 2, 3.

6.1 Introduction to Probability

Definition of terms (Contd. . . .)

- ii. Roll a die once and observe the numbers that turns on top. Here the possible outcomes are 1, 2, 3, 4, 5, 6.



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Rolling a die

6.1 Introduction to Probability

Definition of terms (Contd. . . .)

Definition 3:- The set of all possible outcomes is called a sample space and is denoted by S.

Example 3:- i. Consider example 1(i) of definition 1. The sample space is $S = \{ H, T \}$

ii. Consider example 1(ii) of definition 1.

What is the sample space?

6.1 Introduction to Probability

Definition of terms (Contd. . . .)

iii. Consider example 2(ii) of definition 2. What is the sample space?

Definition 4:- Any subset of the sample space is called an event.

Example 4:- Consider example 2(i). We have the following events for $S = \{0, 1, 2, 3\}$.

6.1 Introduction to Probability

Definition of terms (Contd. . . .)

Let A: The number of heads is greater than 1.

i.e. $A = \{2, 3\}$.

B: The number of heads is less than 3.

i.e. $B = \{0, 1, 2\}$

C: No head turns on top.

i.e. $C = \{0\}$

6.1 Introduction to Probability

Definition of terms (Contd. . . .)

Definition 5:- Two events are said to be *mutually exclusive* if they cannot both occur at the same time.

Example 5:- i. Consider an experiment of rolling a die once. Define events A and B as:

6.1 Introduction to Probability

Definition of terms (Contd. . . .)

A: an odd number turns on top

i.e., $A = \{1, 3, 5\}$



B: an even number turns on top

i.e., $B = \{2, 4, 6\}$.



Thus $A \cap B = \emptyset$, and hence by definition

A and B are mutually exclusive events.

6.1 Introduction to Probability

Definition of terms (Contd. . . .)

- ii. Consider again an experiment of rolling a die once. Define an event C as:

C: a number less than 3 turns on top = $\{1, 2\}$

Then neither A and C nor B and C are mutually exclusive events since $A \cap C \neq \emptyset$ and $B \cap C \neq \emptyset$.

6.1 Introduction to Probability

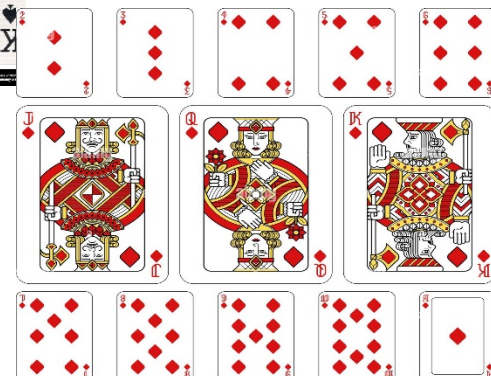
Definition of terms (Contd. . . .)

iii. Consider an experiment of drawing a card once from a well shuffled ordinary deck of 52 cards. Define events A, B, and C as:

A: The drawn card is a Jack.

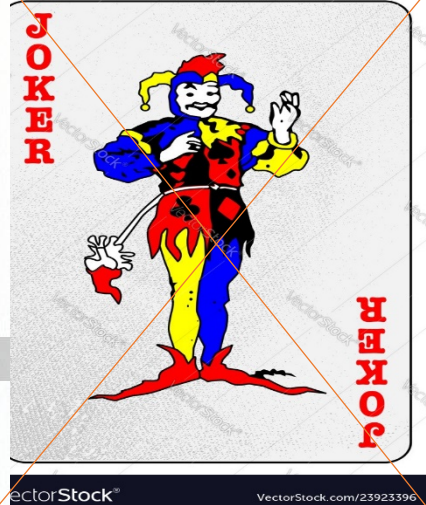
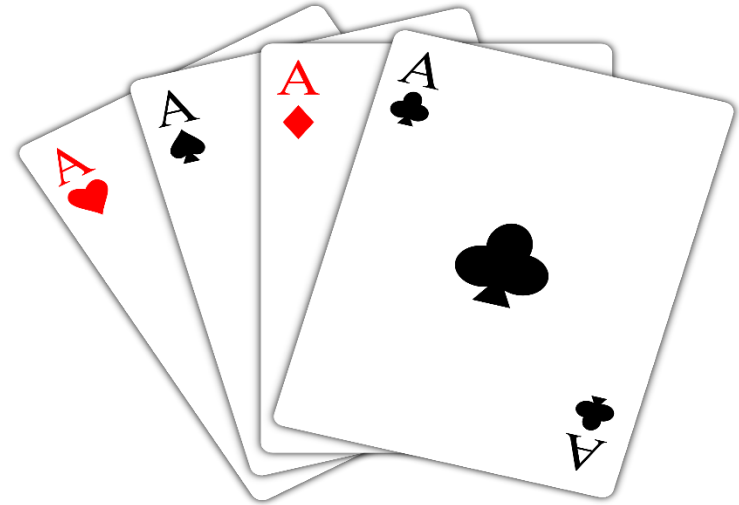
B: The drawn card is a King

C: The drawn card is a red card.



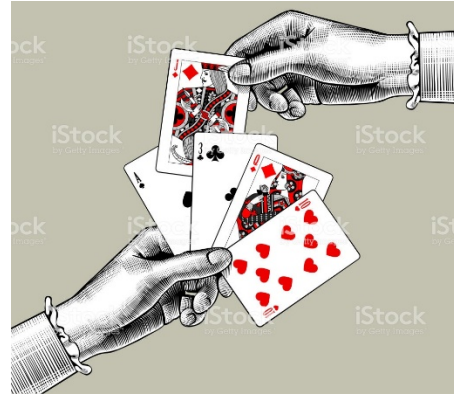
6.1 Introduction to Probability

Definition of terms (Contd. . . .)



6.1 Introduction to Probability

Definition of terms (Contd. . . .)



6.1 Introduction to Probability: Definition of terms (Contd. . . .)

****** *Define two events which are mutually exclusive and which are not mutually exclusive.*

Definition 6:- Suppose an event A can happen in h ways out of a total of n possible equally likely ways. probability of occurrence of event A is defined as:

$$P(A) = \frac{n(A)}{n(S)}$$

6.1 Introduction to Probability: Definition of terms (Contd. . . .)

• *Example 6:-* Consider an experiment of rolling a die once. What is the probability of getting

- | | |
|-------------------|---------------------|
| (i) a 5? | (ii) an odd number? |
| (iii) a 1 or a 4? | (iv) a 7? |

Postulates of probability

6.1 Introduction to Probability: Postulates of Probability

Let E be an experiment for which the sample space is S.

1. For any event A, $0 \leq P(A) \leq 1$.

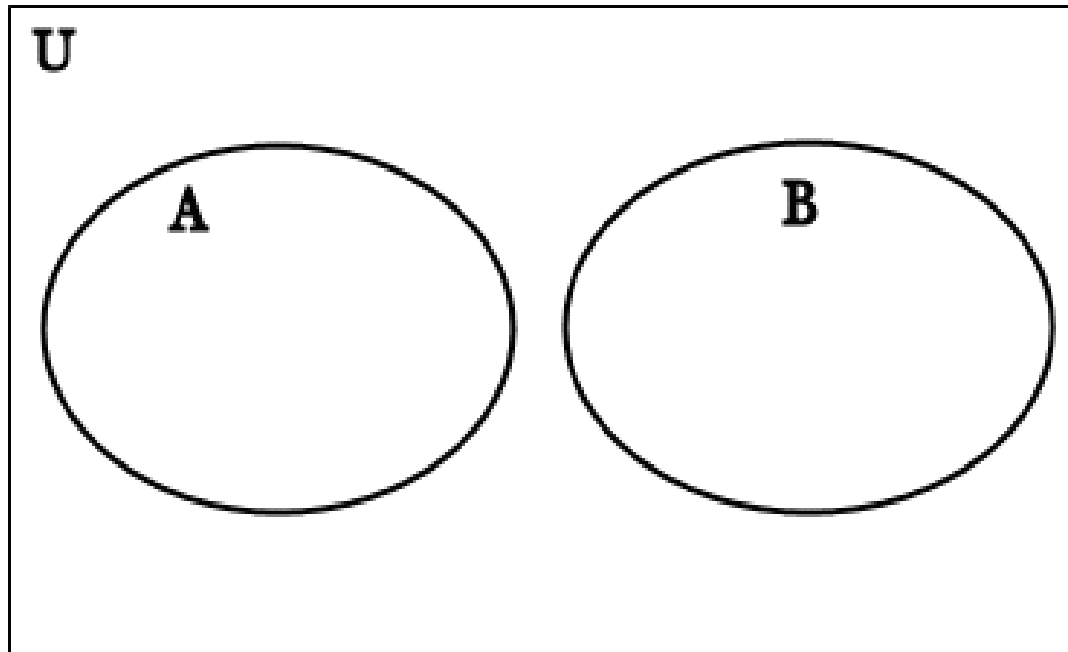
Definition 7:- If the probability of an event is zero, then we refer to the event as null or impossible event; on the other hand if the probability of an event is one, then we refer to the event as *sure or certain* event.

6.1 Introduction to Probability: Postulates of Probability

2. $P(S) = 1.$
3. Addition Rule For Two *Mutually Exclusive Events.*

*If A and B are two mutually exclusive events,
then $P(A \text{ or } B) = P(A \cup B) = P(A) + P(B).$*

6.1 Introduction to Probability: Postulates of Probability



Disjoint sets or Mutually Exclusive events:-

$$A \cap B = \emptyset$$

6.1 Introduction to Probability: Postulates of Probability

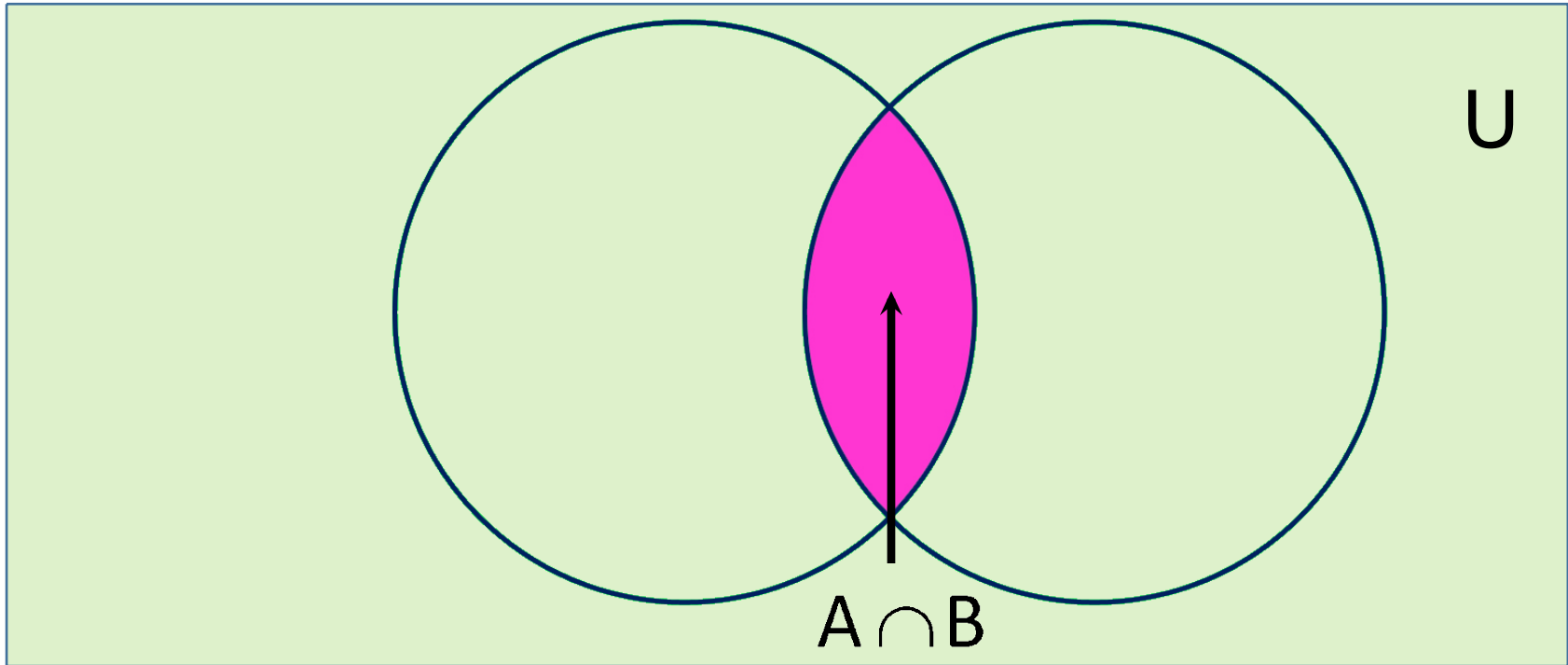
4. Addition Rule for two events that are
not Mutually Exclusive.

If A and B are *not mutually exclusive events*,

then $P(\mathbf{A \text{ or } B}) = P(\mathbf{A}) + P(\mathbf{B}) - P(\mathbf{A \text{ and } B})$

OR $P(\mathbf{A \cup B}) = P(\mathbf{A}) + P(\mathbf{B}) - P(\mathbf{A \cap B}).$

6.1 Introduction to Probability: Postulates of Probability



Non disjoint sets or Non Mutually Exclusive Events:-

$$A \cap B \neq \emptyset$$

6.1 Introduction to Probability

Postulates of Probability

Example 7:- A card is drawn from a well shuffled ordinary deck of 52 cards. What is the probability that the drawn card is:

- (i) a Queen or a Jack?
- (ii) a Queen or a black card?
- (iii) a face (picture) card or a red card
- (iv) a picture card or a diamond?

6.1 Introduction to Probability

Postulates of Probability

Example 8:- In a soft drink bottling company, 5% of the bottles are under filled and 6% of the bottles are improperly capped. It was also known that 2% of the bottles are under filled and at the same time improperly capped. What is the probability of obtaining a bottle having either of the defects?

Any Questions



Any Questions

